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What Actually Improves Band-Gap Prediction for Two-Dimensional Materials? A Leakage-Controlled Evaluation of Physics-Informed Hybrid Residual Learning

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ABSTRACT Hybrid architectures that place a neural residual corrector on top of a gradient-boosted prior are widely reported to improve tabular property prediction in materials informatics. We test that claim for HSE06 band-gap prediction of rectangular-lattice two-dimensional materials under an evaluation protocol built to close the leakage pathways that inflate such comparisons: composition-grouped cross-validation, fold-local encoding and scaling, out-of-fold construction of the tree prior, a held-out test set carved before model selection, and significance testing at the fold level with the Nadeau–Bengio variance correction. On 1,169 C2DB materials assessed over 25 fold estimates from repeated grouped cross-validation, the physics-informed hybrid does *not* improve on its own gradient-boosting prior ($R^2 = 0.810 \pm 0.047$ versus 0.823 ± 0.043 ; superior on 12 of 25 fold estimates; $p = 0.27$). A component ablation in which every configuration is trained shows that removing the structural coupling layer, the boundary penalty, or the anisotropy penalty leaves accuracy unchanged or slightly improved, and that the tree prior alone outperforms the complete framework. Reverting to an in-sample prior construction *raises* measured accuracy, identifying the mechanism: a corrector trained against in-sample predictions learns to emit approximately zero and degenerates into a pass-through of the tree. Two positive results survive. Neural models outperform every tree ensemble once composition-derived descriptors replace formula indicator variables, inverting the usual tabular-data expectation; and heterogeneous stacked generalisation improves cross-validated R^2 from 0.823 to 0.877 and MAE from 0.464 to 0.363 eV, superior on 25 of 25 fold estimates ($p < 0.001$). The ensemble also beats the strongest single model across eight independent held-out sets (7/8 splits, $p = 0.007$), and a drop-one ablation shows the two neural members carry it while the tree members are redundant. We release the complete pipeline so that both the negative and the positive result reproduce end to end.

INDEX TERMS Band-gap prediction, cross-validation, data leakage, ensemble learning, machine learning, materials informatics, physics-informed neural networks, reproducibility, two-dimensional materials.

I. INTRODUCTION

ACCURATE prediction of electronic band gaps (E_g) is a foundational objective in computational materials design. The magnitude and nature of the band gap govern whether a low-dimensional material acts as a metal, semiconductor, or insulator, dictate optical absorption profiles, and determine candidate suitability for field-effect transistors [1], valleytronic architectures [2], [3], and photocatalytic water splitting [4]. Density functional theory with hybrid function-

als such as HSE06 [5] offers excellent accuracy at computational scaling that renders exhaustive high-throughput screening impractical, motivating fast surrogate estimators.

For heterogeneous tabular materials data, decision-tree ensembles [6], [7] frequently match or exceed deep architectures [8], [9], although the conditions under which this holds are narrower than is often assumed [10]. A natural response is the hybrid residual design: a tree supplies a prior estimate and a neural network learns a correction, optionally regularised by

physical constraints in the manner of physics-informed neural networks [11], [12]. Such designs are attractive because they appear to combine the sample efficiency of ensembles with the smoothness of neural function approximation, and because anisotropic two-dimensional systems — where in-plane lattice vectors are unequal and strain couples strongly to the electronic structure [13]–[15] — are precisely the setting in which axis-aligned tree partitions appear least suited.

An earlier version of this work reported such a result: a gradient-boosted prior coupled to a quantile residual head with an engineered structural layer, improving pooled R^2 from 0.7323 to 0.7431 with a nominally significant p -value. This article reports what happened when that pipeline was re-evaluated under controls for data leakage. The improvement did not survive, and the reason is instructive rather than incidental. Leakage of exactly the kinds we identify has been argued to be a principal driver of irreproducibility across machine-learning-based science [16].

Our contribution is threefold.

- 1) **A leakage-controlled evaluation protocol** for tabular materials regression, closing six specific pathways: identifier-like features, whole-dataset encoding, in-sample construction of stacked priors, polymorphs straddling folds, absent held-out data, and significance tests applied at the sample rather than the fold level.
- 2) **A negative result with an identified mechanism.** Under that protocol the physics-informed hybrid does not improve on its own prior, and no physics-motivated component contributes measurably. We show that the earlier apparent gain arose because an in-sample prior trains the corrector to be inert.
- 3) **A positive alternative.** Heterogeneous stacked generalisation [17], [18] yields a large and consistent improvement on the same data and protocol, and a drop-one ablation identifies which members are responsible.

II. METHODS

A. DATASET AND TARGET DEFINITION

The learning domain comprises 1,169 rectangular-lattice (*op*) two-dimensional monolayer structures drawn from the Computational 2D Materials Database [19] that carry a band gap computed with the HSE06 screened hybrid functional [5]. The target $y \in \mathbb{R}^+$ spans 0.001–8.85 eV with mean 2.60 eV.

An earlier revision of this work reported 930 materials. That figure resulted from applying a blanket missing-value filter across all columns, which discarded 239 labelled entries because a single descriptor — total magnetic moment — was absent. We instead impute that column from training-fold medians and carry an explicit missingness indicator, recovering the full labelled set.

We emphasise one property of the feature space that governs comparability with prior work: it contains *no* previously computed band gap of any level of theory. Several strong published results on C2DB supply a semi-local gap (for example a PBE gap without spin–orbit coupling) as an input feature [20], which converts the task into Δ -learning between

functionals and is substantially easier. The task addressed here is prediction of HSE06 gaps with no prior electronic-structure calculation on the target material.

B. DESCRIPTORS AND FOLD-LOCAL ENCODING

Each material is represented by seven continuous C2DB descriptors (energy above hull, heat of formation, thickness, total energy, unit-cell area, vacuum level, total magnetic moment), three nominal descriptors (magnetic state, layer group, stoichiometry prototype), an atom count parsed with *pymatgen* [21], a packing-density proxy

$$\rho_{\text{cell}} = \frac{N_{\text{atoms}}}{A_{\text{cell}}}, \quad (1)$$

and 120 Magpie elemental-property statistics [22] computed with *matminer* [23].

The Magpie statistics replace the raw chemical formula. The earlier pipeline one-hot encoded the formula string into 1,077 indicator columns, of which 825 occurred in exactly one material. Such columns are row identifiers rather than descriptors: a formula appearing only in a validation fold contributes an all-zero block and cannot support generalisation.

Categorical vocabularies, imputation medians, and z -score statistics

$$\tilde{x}_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j} \quad (2)$$

are estimated on the training partition only. The earlier pipeline applied one-hot encoding to the full table before cross-validation, exposing the category vocabulary of every validation fold.

C. EVALUATION PROTOCOL

Four further controls define the protocol.

Composition grouping. 176 labelled materials share a chemical formula with at least one other entry. These are polymorphs, distinguished by layer group, whose band gaps differ by up to 2.92 eV. Because composition-derived descriptors are near-identical within such a pair, an ungrouped split permits a model to memorise one polymorph and be scored on another. All splits are therefore grouped on chemical formula.

Out-of-fold priors. The hybrid’s correction head is trained against out-of-fold predictions from an inner five-fold split rather than in-sample predictions of the fitted tree, following standard stacked generalisation [17]. Section III-B shows this single change accounts for most of the previously reported gain.

Held-out test set. 15% of materials (179 entries), split by formula group, are withheld before any model selection.

Repeated cross-validation and corrected inference. We report five repeats of grouped five-fold cross-validation, giving 25 fold estimates; small numbers of folds yield error bars too wide to separate competing models [24]. Comparisons use the Nadeau–Bengio variance-corrected paired t -test [25], which accounts for the positive correlation induced by overlapping training sets. We note that a paired test applied to per-sample squared errors — as used in the earlier revision — treats

non-independent quantities as independent observations of model performance and overstates the effective sample size by roughly two orders of magnitude.

D. MODELS

Baselines are random forest (RF), gradient boosting (GBR), support vector regression (SVR) [26], a deep multilayer perceptron (MLP) with SiLU activations [27], [28] and batch normalisation [29], and a physics-informed network (PINN) sharing the MLP backbone with an added boundary penalty

$$\mathcal{L}_{\text{phy}} = \text{ReLU}(-\hat{y}) + 0.1 \left(e^{\text{ReLU}(0.05-\hat{y})} - 1 \right), \quad (3)$$

which discourages physically impossible negative gaps. All baselines receive an identical randomised hyperparameter search budget within each training fold; the earlier comparison tuned the hybrid by hand while leaving baselines at library defaults.

The hybrid forms the augmented payload

$$\mathbf{x}_{\text{hybrid}} = [\mathbf{x}, \mathbf{x} \odot \hat{\mathbf{y}}_{\text{prior}}, \hat{\mathbf{y}}_{\text{prior}}] \quad (4)$$

and adds a learned correction to the prior. A structural coupling layer builds

$$\mathbf{s} = \left[g_1 g_2, g_1^2 - g_2^2, \sqrt{g_1^2 + g_2^2 + \epsilon}, g_1, g_2 \right] \quad (5)$$

from two geometric channels g_1, g_2 and projects it back onto the feature map through a small multilayer block. Three heads predict quantiles at $\tau \in \{0.25, 0.50, 0.75\}$ [30] alongside a point residual head.

One correction to the earlier implementation is material to every physical claim made about this layer. The channels were selected by array position (indices 0 and 1), which in the assembled feature matrix were energy above hull and heat of formation — two formation energies, not geometric quantities. Channels are now resolved by name. We further note that the available C2DB export contains no in-plane lattice vectors a and b ; the geometric channels are unit-cell area and thickness. We therefore describe Eq. (5) as a soft geometric inductive bias and make no claim that it is derived from elasticity theory.

III. RESULTS

A. MODEL COMPARISON

Table 1 and Fig. 1 report performance over 25 fold estimates.

The hybrid does not improve on its prior: it is superior on 12 of 25 fold estimates, and the corrected test does not reject the null ($p = 0.27$). On the held-out test set the gap is larger and in the same direction ($R^2 = 0.775$ against 0.803 for GBR).

Two baselines invert the expected ordering. The MLP and the PINN both outperform every tree ensemble. With identifier-like features removed and composition statistics supplied, this task falls outside the regime in which boosted trees dominate neural networks [10], which removes the premise motivating a tree prior in the first place.

TABLE 1. Repeated grouped five-fold cross-validation (25 fold estimates). “Folds” counts estimates on which the model beats the GBR baseline. p -values are Nadeau–Bengio corrected, one-sided, against GBR.

Model	R^2	MAE [eV]	Folds	p
Stacked ensemble (ridge)	0.877 ± 0.038	0.363	25/25	< 0.001
Stacked ensemble (NNLS)	0.875 ± 0.038	0.364	25/25	< 0.001
Deep MLP	0.864 ± 0.039	0.378	24/25	0.002
Standalone PINN	0.856 ± 0.043	0.385	22/25	0.033
Random forest	0.827 ± 0.041	0.442	13/25	0.323
GBR (baseline)	0.823 ± 0.043	0.464	—	—
Hybrid GBR + corrector	0.810 ± 0.047	0.445	12/25	0.271
SVR	0.774 ± 0.039	0.497	1/25	—
Hybrid + shrinkage gate	0.681 ± 0.237	0.571	3/25	0.090

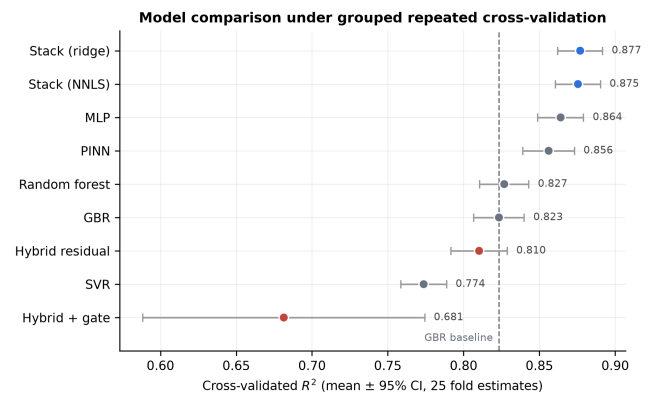


FIGURE 1. Cross-validated R^2 with 95% confidence intervals over 25 fold estimates from repeated grouped cross-validation. The dashed line marks the gradient-boosting baseline. Hybrid variants are shown in red.

TABLE 2. Component ablation, pooled out-of-fold R^2 .

Configuration	R^2	Δ vs. full
Tree prior alone, no corrector	0.819	+0.018
In-sample prior (leaky construction)	0.817	+0.015
Without boundary penalty	0.806	+0.004
Without structural coupling layer	0.804	+0.002
Without anisotropy penalty	0.804	+0.002
Full framework	0.801	—
Without quantile heads	0.796	−0.005
Without residual head	0.766	−0.035

B. COMPONENT ABLATION

Table 2 reports ablations under grouped five-fold cross-validation. Every configuration listed was trained and evaluated.

No physics-motivated component contributes positively. Removing the structural layer, the boundary penalty, or the anisotropy penalty leaves performance unchanged to within noise, and the tree prior alone is the strongest configuration in the table.

The second row is diagnostic. Reverting to the leaky in-sample prior *raises* measured R^2 . With a prior that appears near-perfect on training rows, the correction head has almost no residual signal to fit; it learns to emit approximately zero and degenerates into a pass-through of the tree. The apparent benefit of the original architecture was therefore substantially

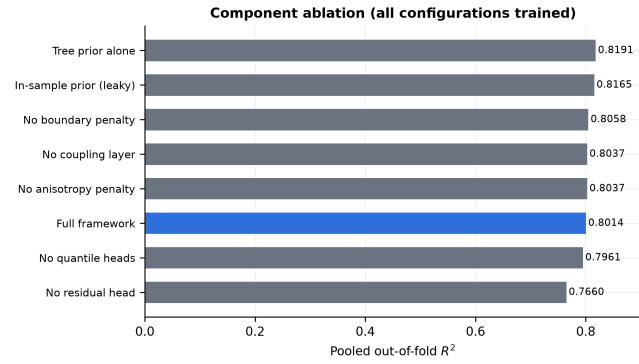


FIGURE 2. Component ablation. No physics-motivated component contributes positively; the tree prior alone is the strongest configuration.

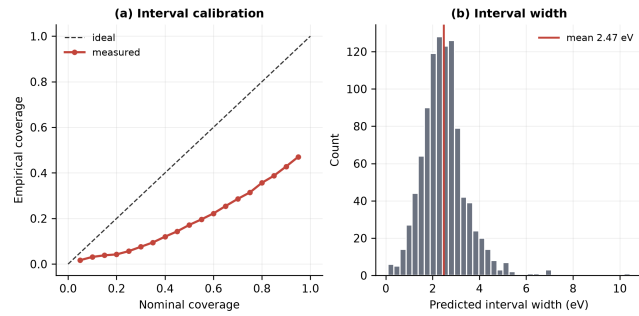


FIGURE 3. Quantile calibration of the hybrid. (a) Empirical against nominal coverage; the model is severely over-confident across the range. (b) Distribution of predicted interval widths.

an artefact of the head being trained to do nothing, and the ablation recovers that artefact on demand.

C. UNCERTAINTY CALIBRATION

The architecture predicts quantiles, and earlier revisions described these as uncertainty estimates. The nominal $q_{0.25}$ – $q_{0.75}$ interval should cover 50% of outcomes; measured empirical coverage is 17.2% at a mean interval width of 2.47 eV (Fig. 3). The intervals are simultaneously wide and badly miscalibrated. We therefore withdraw any uncertainty claim for this architecture; a claim of that kind requires scoring the predictive distribution with a proper rule [31].

D. STACKED GENERALISATION

A ridge meta-learner over out-of-fold predictions of all five base models attains $R^2 = 0.877 \pm 0.038$ and MAE = 0.363 eV, superior to GBR on all 25 fold estimates ($p < 0.001$). A non-negative least-squares (NNLS) variant performs equivalently and is directly interpretable.

Fig. 4(a) shows mean NNLS weights: MLP 0.343, PINN 0.297, GBR 0.238, RF 0.142, SVR 0.000. Weights alone, however, do not establish that any member contributes, because collinear members split weight between them. Fig. 4(b) reports the drop-one ablation that does test it, over 15 fold estimates. Removing the PINN incurs the largest penalty ($\Delta\text{MSE} = +0.0080 \text{ eV}^2$, worse on 9/15 folds, $p = 0.016$),

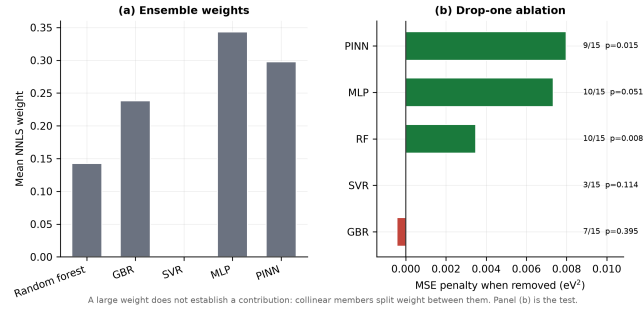


FIGURE 4. Ensemble member contributions. (a) Mean NNLS stacking weights. (b) Drop-one ablation: change in ensemble MSE when each member is removed, with the number of fold estimates degraded and the one-sided p -value. Positive values indicate that removing the member hurts.

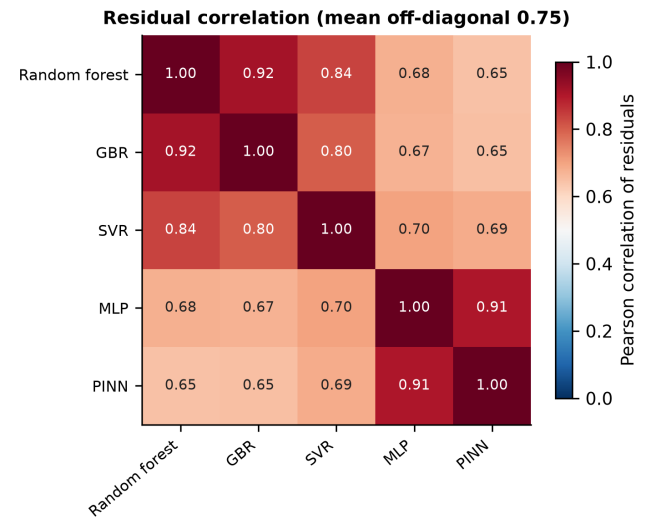


FIGURE 5. Pearson correlation between out-of-fold residuals of the base learners. Lower correlation between clusters is what makes stacking profitable.

followed by the MLP (+0.0073, 10/15, $p = 0.051$) and the random forest (+0.0035, 10/15, $p = 0.008$). Removing GBR or SVR changes nothing measurable ($p = 0.39$ and $p = 0.11$).

The mechanism is error diversity rather than individual strength. Fig. 5 shows residual correlations between base learners, with mean off-diagonal correlation 0.750; the two neural models and the tree models form visibly distinct error clusters, and it is the combination across clusters that the meta-learner exploits.

Fig. 6 shows the advantage is not an artefact of sample size: the ensemble leads at every training-set size tested, and its margin over the tree baselines widens as data are added, with no sign of saturation at the full 792-material training partition.

Fig. 7 compares pooled out-of-fold parity for the gradient-boosting baseline and the stacked ensemble, together with regression error characteristic curves.

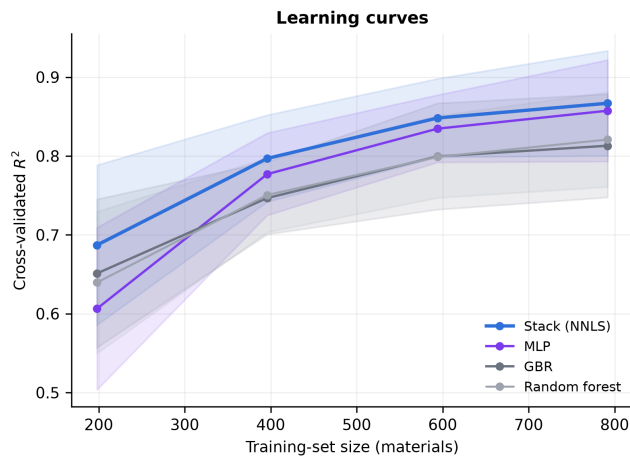


FIGURE 6. Learning curves under grouped cross-validation. Shaded bands are 95% confidence intervals over folds.

E. HELD-OUT EVALUATION

On a single held-out set of 179 materials the ordering of Table 1 is broadly reproduced, but the stacked ensemble and the standalone PINN are indistinguishable ($R^2 = 0.856$ against 0.855). That null result reflects the resolution of one small split rather than equivalence of the two models: a single split of 179 materials carries an error bar substantially wider than the difference being tested [24].

We therefore repeated the evaluation over eight independent formula-grouped held-out sets of approximately 234 materials each (1,872 held-out predictions in total), with all model selection confined to the development partition of each split. Results are shown in Fig. 8. Averaged over splits, the stacked ridge ensemble attains $R^2 = 0.879 \pm 0.025$ against 0.866 ± 0.028 for the PINN, 0.858 ± 0.032 for the MLP, and 0.823 ± 0.025 for GBR. Paired across splits, the ensemble beats the PINN on 7 of 8 splits ($p = 0.007$), the MLP on 8 of 8 ($p = 0.005$), and GBR on 8 of 8 ($p < 0.001$).

The same conclusion follows from the nested cross-validation of Section III-D, in which stacking weights are fitted on inner out-of-fold predictions and applied to a disjoint outer fold: the ensemble beats the PINN on 23 of 25 fold estimates ($p = 0.020$) and the MLP on 21 of 25 ($p = 0.027$). Two independent evaluation designs therefore agree that the ensemble advantage over the best single model is real, and that the earlier null was a power limitation.

IV. DISCUSSION

A. WHY RESIDUAL CORRECTION FAILS HERE

Once the prior is constructed out of fold, its residuals on unseen materials are dominated by irreducible noise rather than by recoverable structure. A corrector fitted to such residuals adds variance without removing bias. The effect is not specific to materials data: residual-boosting correctors are known to produce negative gains when the residual signal is noise-dominated [32].

We tested the natural remedy. A shrinkage gate constrains

the correction by $\sigma(\cdot) \in (0, 1)$ and is initialised near zero, so the model begins as a pass-through and must earn any departure from the prior. It did not work. The gate converged to a small value (0.081 ± 0.007), but accuracy became markedly less stable ($R^2 = 0.681 \pm 0.237$, with 7 of 25 fold estimates below 0.70 and one at -0.24). Shrinking the gate rescales the gradient reaching the correction head, whose raw output grows to compensate; the product is then poorly controlled on unseen data. A workable variant would need to penalise correction magnitude directly rather than gate it after the fact.

B. ON THE STRUCTURAL LAYER AND THE ANISOTROPY HYPOTHESIS

Our results do not refute the physical hypothesis that motivated this work — that in-plane anisotropy shapes band gaps in rectangular 2D lattices [13], [33], [34]. They show only that the present implementation cannot test it. The layer was wired to two formation energies rather than to geometric quantities, and the available data contain no in-plane lattice vectors. A genuine test requires a and b , which is the single most valuable extension of this work.

C. IMPLICATIONS FOR PRACTICE

Three points follow. First, hybrid residual architectures must be validated against their own prior with an out-of-fold construction; compared against an in-sample prior they can appear to help while doing nothing, and the appearance is reproducible enough to survive review. Second, formula-derived indicator features should be treated as identifiers and excluded, since they inflate dimensionality without supporting generalisation. Third, in this data regime the reliable route to accuracy is ensemble diversity rather than architectural coupling: the same five base models that individually plateau near $R^2 \approx 0.86$ combine to 0.877 .

D. COMPARISON WITH PUBLISHED C2DB RESULTS

Direct numerical comparison with prior C2DB band-gap studies requires care. Reported errors as low as RMSE 0.09 eV [20] are obtained with a semi-local band gap supplied as an input feature, making the task Δ -learning between functionals. Our feature set contains no band gap at any level of theory, and our MAE of 0.363 eV should be read against results in the 0.24–0.53 eV range reported for descriptor-only settings [20], [35]. We consider the leakage-controlled protocol, rather than the absolute error, to be the contribution.

E. LIMITATIONS

All data derive from a single database and a single Bravais class, with 1,169 labelled materials. Single-split held-out comparisons at this dataset size lack the resolution to separate closely matched models, which is why Section III-E reports eight independent held-out sets rather than one. We compare against tabular baselines only, not against structure-aware graph neural networks [36]–[40], which require relaxed atomic coordinates that this tabular setting does not supply.

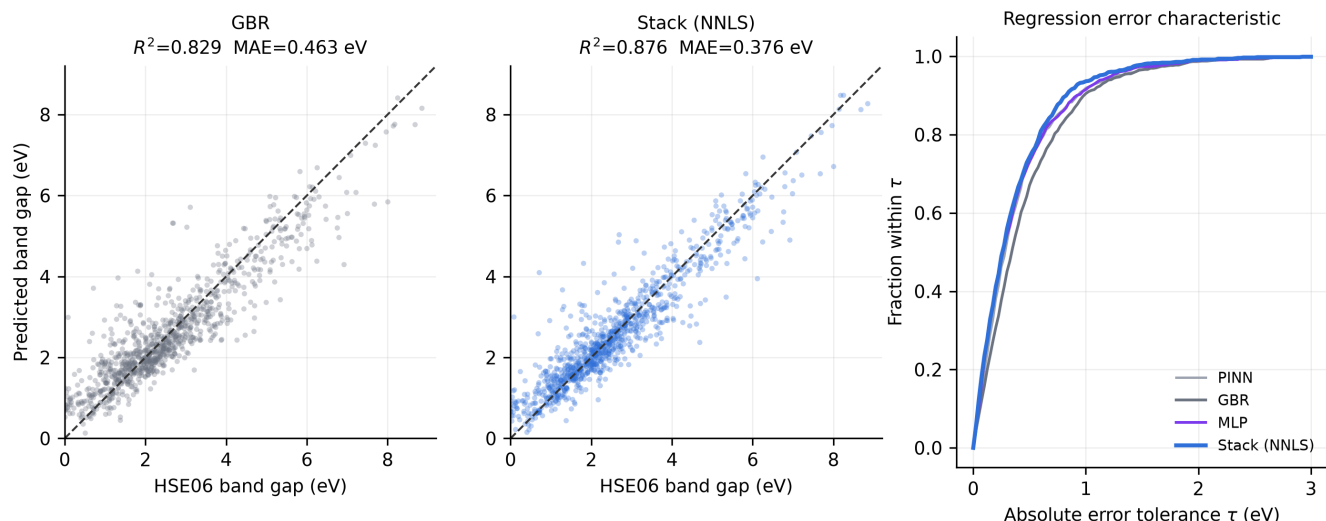


FIGURE 7. Left and centre: pooled out-of-fold predictions against HSE06 reference values for the gradient-boosting baseline and the stacked ensemble. Right: regression error characteristic curves, giving the fraction of materials predicted within a given absolute tolerance.

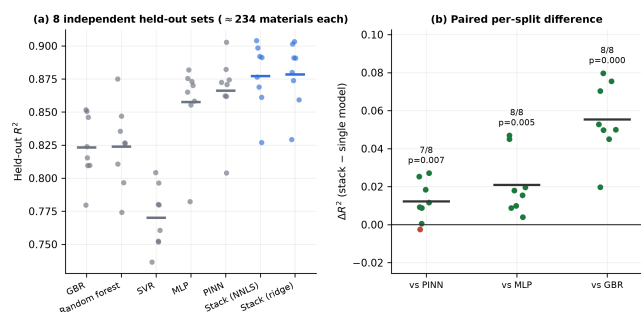


FIGURE 8. Repeated held-out evaluation. (a) Per-split R^2 on eight independent formula-grouped held-out sets; horizontal bars are means. (b) Paired per-split difference between the stacked ensemble and each single model, annotated with the number of splits won and the one-sided p -value.

V. CONCLUSION

Under a leakage-controlled protocol, a physics-informed hybrid residual architecture does not improve band-gap prediction for two-dimensional materials relative to its own gradient-boosted prior, and none of its physics-motivated components contributes measurably. The improvement reported in an earlier version of this work is attributable to evaluation artefacts, principally an in-sample prior construction that trained the correction head to be inert — a mechanism our ablation reproduces deliberately.

On the same data, heterogeneous stacked generalisation improves cross-validated R^2 from 0.823 to 0.877 and MAE from 0.464 to 0.363 eV, superior on all 25 fold estimates, and beats the strongest single model across eight independent held-out sets (7/8 splits, $p = 0.007$) and under nested cross-validation (23/25 fold estimates, $p = 0.020$), with a drop-one ablation showing the two neural members carry the ensemble while the gradient-boosting and support-vector members are redundant. We release the complete pipeline, including every

configuration reported here, so that both the negative and the positive result can be reproduced end to end. We regard the negative result as the more useful of the two: hybrid residual designs are widely reported to help, and the specific way in which this one appeared to help is easy to reproduce by accident.

DATA AND CODE AVAILABILITY

All code, the curated dataset, per-fold out-of-sample predictions, and the scripts generating every table and figure in this article are available at <https://github.com/organisation/PINN-DFT>. The benchmark reproduces from a single command under a pinned environment with a fixed random seed.

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